HEALTH-A-THON 2026 · CANCER TRACK · CLINIC OPERATIONS & PATIENT FLOW

RAD-SMART

The right patient, on the right machine, at a predictable time.

Radiotherapy AI-assisted Dynamic Scheduling, Machine Allocation and Resource Tracking



One planned treatment day: each block is a patient's session (routine, new start, complex, blood irradiation)

Dr Akshay Dinesan, Manipal: team leader and doctor partner · **Abhinand T M**: technical lead · **P S Raghotham Rao**: lead engineer

Dr Shirley Lewis Salins: doctor partner · **Dr Umesh Velu**: doctor partner

Assistive, not diagnostic

THE PROBLEM

One machine, 70–90 patients a day, long waits

1 in 10

patients waits more than **1 hour 49 min** for a daily session

22%

are treated more than an hour after their appointment

The wait grows through the day

Typical (median) wait, by when the patient arrives



Times are given by hand, without knowing how long each session takes, so too many patients are told to come at once.

A course lasts 5–7 weeks, every weekday, usually with a family member. 69% of India's radiotherapy centres have only one machine.

Department's records: 55 days, Oct–Dec 2024, 3,137 patient visits (anonymised, totals only) · J Med Phys 2024

WHY IT IS HARD

It is not just booking appointments

Sessions differ

A routine session takes 7–15 minutes; a new patient or a complex case takes 20–60.

Rules must hold

New patients before 17:00 · hospice patients back for the 17:00 bus · bus users home by 21:00 · one breast board, shared

The day changes

Breakdowns of 15 minutes to 48 hours, urgent patients, late arrivals and no-shows.

Each patient keeps **one time slot for the whole course**. A good plan must find that slot, keep every rule, and repair the day in minutes when something goes wrong.

Rules confirmed by the department, 21 Sep and 24 Oct 2026

OUR CHOICE

We compared five ways; a hybrid scored best



Score out of 5 for safety, impact, speed, ease of building, explaining and languages. A chatbot alone cannot promise the rules; a solver alone cannot learn or talk to patients. The hybrid uses each where it is strongest.

Weights: safety 20%, impact 20%, real-time 15%, feasibility 15%, explainability 10%, languages 10%, configurability 10%

THE SOLUTION

Four steps, and a person decides

1 · PREDICT

How many minutes each patient needs on the machine

2 · PLAN

One time slot per course, with every rule guaranteed

3 · TEST

Try the plan in a virtual copy of the department first

4 · TELL

Explain and message patients in Kannada, Tulu and more

PEOPLE DECIDE

The doctor approves each course. A technologist or doctor approves every change on the day. Nothing reaches a patient before that.

It never decides *whether* or *how* a patient is treated, only *when*. The language model explains; it never schedules.

WORKING APP, INSIDE CARE

One button on the patient's page

DOCTOR

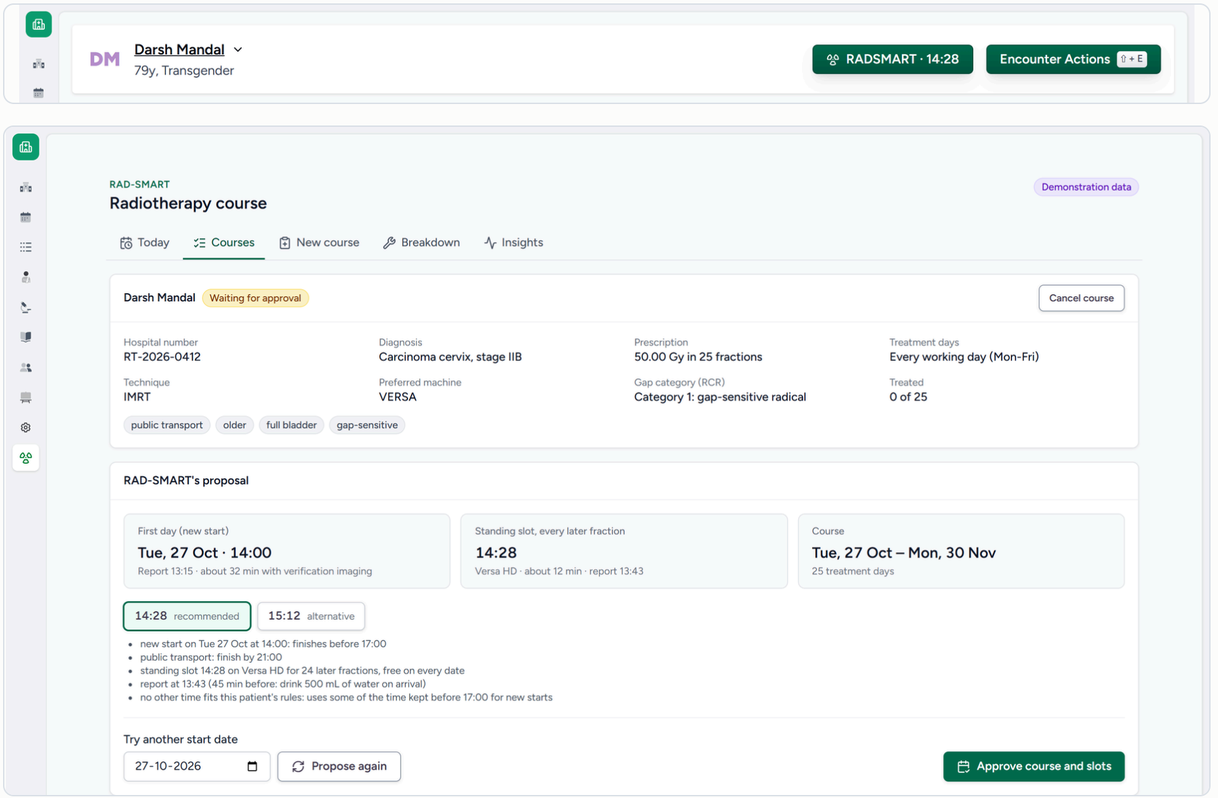
Clicks **RADSMART**, enters the course, approves the proposed time

TECHNOLOGIST

Runs the day: who has arrived, who is in the room, machine status

BREAKDOWN

Enters the downtime; RAD-SMART proposes the new day for approval



RESULTS · WITH AND WITHOUT RAD-SMART

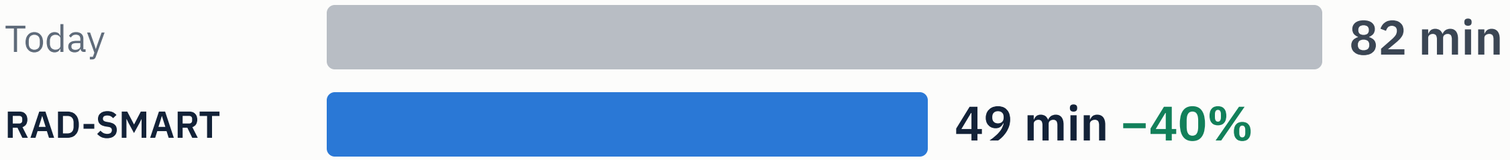
Long waits fall by more than a third

The long wait: 1 patient in 10 waits longer than this

The department's own 55 days, replayed



12 simulated weeks, one slot per course



22% → 6%
treated more than an hour late

130 → 70 min
on the busiest days, the biggest gain

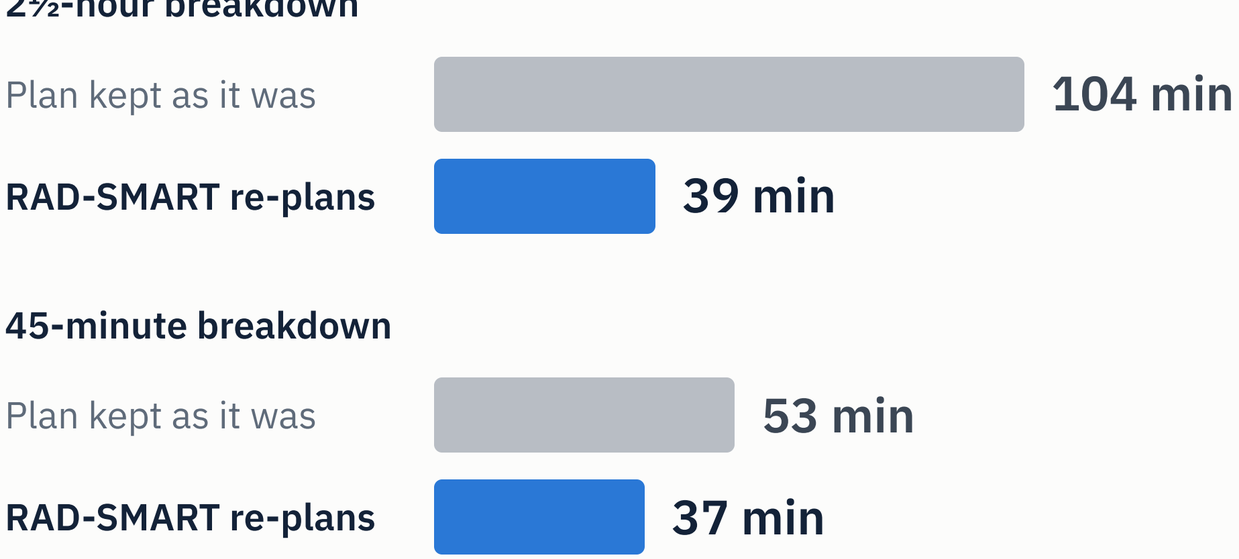
The trade-off
the evening runs about 2 hours later

Wait from arrival to the treatment room · replayed in a virtual copy of the department, not yet a live trial

WHEN THE MACHINE BREAKS DOWN

A new plan in seconds, not a lost day

Typical wait, arrival to treatment



A new plan in 10–14 seconds; patients still at home get their new time by message.

THE DEPARTMENT'S RULES

Up to 1 hour: nobody is sent home; everyone is treated, with overtime

Longer: only as many patients are sent back as needed, those still at home first

Always: a technologist or doctor approves first

150-minute breakdown at 10:00, 300 simulated days, synthetic patients · rules of 24 Oct 2026

NEXT STEP

Safe by design, then a 90-day pilot

Days 1–14
Measure
Record today's times

Days 15–35
Shadow
RAD-SMART plans; staff compare; nothing changes

Days 36–90
Live
Doctors and technologists use it in CARE; patients get their times by message

PILOT TARGET

25% shorter

long waits (today 109 min), and no rule broken

SAFE BY DESIGN

- A person approves every change
- Rules re-checked on every plan
- No names or numbers sent to AI
- Every change logged

Our ask: a pilot slot with mentors, Sarvam AI credits, and introductions to cancer centres. Cost about ₹2–3 lakh.